

HW 3

Deadline: 4.24.2025 (11:59 PM) (Total points: 100)

You will again design a face recognition system in this HW. Consider the two datasets “GallerySet.rar” and “ProbeSet.rar” provided for Project, where gallery images and probe images are stored.

Background: A gallery image is the image used for enrolling an individual for the first time into a system and a probe image is the image of an individual with which he tries to access the system. The basic working principle of a face recognition system is very simple, i.e., all the genuine users are enrolled into a system in the enrollment step with their gallery images. During the recognition step, an individual tries to access the system by submitting a face template (=probe image) which is matched against all the face templates from the gallery set collected during the enrollment phase. The individual is recognized/authorized when a match is found for him with a high match score. Note that the probe image can be coming from a genuine user or an unauthorized individual who attempts to get authenticated into the system.

Evaluation Metric: There are several performance metrics available to evaluate the performance of a face recognition system. A decidability index value (d') is one of them which will be considered for this HW. It quantifies the separation between the distribution of genuine scores and imposter scores. A score is called genuine if it is produced by comparing a probe template with a gallery template where both the templates belong to the same individual. On the other hand, an imposter score is produced when a probe template is compared to a gallery template where the templates belong to two different individuals.

Tasks Details for the HW: For this HW, the GallerySet contains face images of 100 users and the ProbeSet contains 100 different images from the same individuals. You will build the face recognition system: “System_A” which authenticates a user by using “correlation” properties comparing his gallery and probe image. You will develop the face recognition system and evaluate its performance by following the given instructions:

- 1) In this step, you will build “System_A”. Both the GallerySet and ProbeSet contain gray-toned 3 channel images. Read them as single channel grayscale image. Now considering the given correlation function (see Appendix) as your similarity measure, perform the comparison between each probe template with all the gallery templates and generate a score matrix. Your score matrix A will be a 100x100 matrix where $A[i,j]$ denotes the similarity score generated by comparing the i -th probe with the j -th gallery data. Provide a snippet of the $A[0:9,0:9]$ matrix. (30 points)
- 2) What is the lowest genuine score, and highest impostor score you found from your score matrix? Do you think you can pick a threshold from the entire score matrix for this system which will not cause any false positives or false negatives? Explain your answer. (5+15 = 20 points)
- 3) Using the genuine and imposter scores from the whole score matrix from the previous step, determine the d' using the given formula. Do you think a higher value of d' shows better recognition performance or a lower one will show better performance? Please justify. (20 + 10 = 30 points)

$$d' = \frac{\sqrt{2} |\mu_1 - \mu_0|}{\sqrt{\sigma_1^2 + \sigma_0^2}},$$

Note: μ_1 and μ_0 are the mean of the genuine and imposter score distribution respectively. Similarly, σ_1 and σ_0 denote the standard deviation of the genuine and imposter distribution respectively.

- 4) In step 1 you considered single channel images. Imagine if you considered the images with all three channels, do you think it would add any improvement in the performance of the system? Why or why not, explain. (20 points)

Appendix for HW:

The function is:

```
def normalizedCorrelationCoefficient(img1: np.ndarray, img2: np.ndarray):
    x = img1.reshape((-1,1))
    y = img2.reshape((-1,1))
    xn = x - np.mean(x)
    yn = y - np.mean(y)
    r = (np.sum(xn * yn)) / (np.sqrt(np.sum(xn**2)) * np.sqrt(np.sum(yn**2)))
    return r
```

Background about the function: The normalized correlation coefficient is a commonly used measure for template matching. The coefficient (r) represents the statistical measure of the strength of a linear relationship between two variables. The formula is:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

$\bar{y}$ = mean of the values of the y-variable

Instructions on Deliverables:

- There will be two deliverables for this submission: a PDF and a .py file.
- The PDF should have the answers to the questions and of course the codes pasted as text.
- No zipped submission will be allowed.